

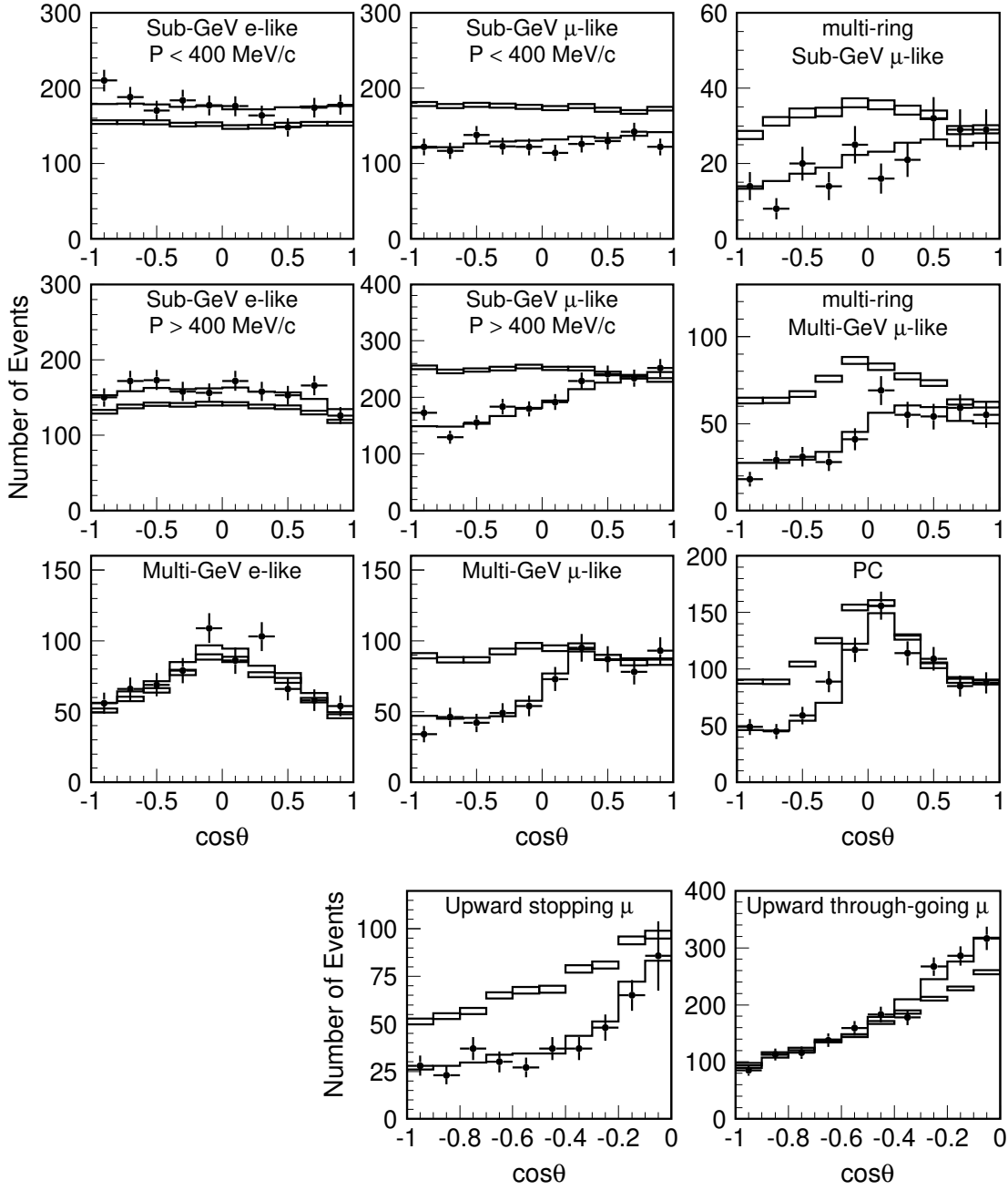


FIG. 31: The zenith angle distribution for fully-contained 1-ring events, multi-ring events, partially-contained events and upward muons. The points show the data, box histograms show the non-oscillated Monte Carlo events and the lines show the best-fit expectations for $\nu_\mu \leftrightarrow \nu_\tau$ oscillations with $\sin^2 2\theta = 1.00$ and $\Delta m^2 = 2.1 \times 10^{-3} \text{ eV}^2$. The best-fit expectation is corrected by the 39 systematic error terms, while the correction is not made for the non-oscillated Monte Carlo events. The height of the boxes shows the statistical error of the Monte Carlo.

ergy scale uncertainty leads to a $^{+0.9}_{-1.1}\%$ error in the stopping muons due to the 1.6 GeV/c cut; the reduction efficiency for stopping (through-going) muons has an uncertainty of $^{+0.34}_{-1.25}\%$ ($^{+0.32}_{-0.54}\%$); and stopping/through-going separation $^{+0.29}_{-0.38}\%$ (where “+” means through-going muons misidentified as stopping). As in the contained event analysis, comparison of data and expectations is done between observed number of events and the live-time-scaled MC number of events. However, to facilitate comparisons with other exper-

iments, these numbers are also presented in units of flux as described in [3, 4]. The additional systematic uncertainty in the observed through-going (stopping) flux comes from effective area of 0.3% and the live-time calculation (0.1%). The absolute expected flux has theoretical uncertainties of at least 20% in the normalization for high energy ($> 100 \text{ GeV}$) neutrinos and 5 to 10% from interaction model differences.

The zenith angle distributions of the upward through-going and stopping muons are shown in Fig. 31. The shape of